DAY 2 | QUANTITATIVE APTITUDE

PERCENTAGES

Beginner to Placement Level — TCS NQT & Service-Based Companies

Covers: TCS NQT, Accenture, Capgemini, Cognizant, Infosys, Wipro, Deloitte, EY, LTIMindtree, Tech Mahindra

Format: Classroom Lecture Notes — 3 Hour Session

Includes: Formula Sheet · Concepts · Shortcuts · 20 Solved Examples · 30 Practice Questions · Answer Key · Revision Sheet

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Section 1 — Introduction to Percentages

1.1 What is a Percentage?

The word "percent" comes from the Latin *per centum*, meaning "per hundred." A percentage is simply a way of expressing a number as a fraction of 100. When we say 40%, we mean 40 out of every 100 parts.

> Core Definition
>
> **$x\% = x/100$**

Percentages let us compare quantities of different sizes on a common scale. Instead of saying "35 out of 70 students passed," we say "50% of students passed" — this is instantly comparable to any other class, of any size.

1.2 Why Percentages Matter for Placement Exams

Percentage is the single most foundational topic in Quantitative Aptitude. Nearly every other topic in TCS NQT and similar exams is built directly on top of it:

- **Profit & Loss** — profit%, loss%, markup%, discount% are all percentage applications.
- **Simple & Compound Interest** — rate of interest is a percentage; CI growth uses successive percentage change.
- **Data Interpretation (DI)** — bar graphs, pie charts, and tables are read almost entirely using percentage change and percentage share.
- **Ratio & Proportion, Averages, Mixtures** — frequently combined with percentage in mixed problems.
- **Time & Work, Speed-Distance-Time** — efficiency and change problems are often phrased as percentage increase/decrease.

> **Placement Insight:** In TCS NQT-style papers, 15–20% of the entire Quant section either directly tests percentages or requires a percentage calculation as an intermediate step. Mastering this topic gives you leverage across the whole test.

1.3 Real-Life Applications

Context	How Percentage is Used
Banking & Finance	Interest rates, EMI calculations, loan markups
Retail & Shopping	Discounts, GST/tax, seasonal sale offers

Exams & Academics	Marks scored, attendance percentage, pass/fail cut-offs
Business	Profit margins, growth rate, market share
Demographics	Population growth/decline rate, literacy rate
Elections	Vote share, winning margin

1.4 Percentage ↔ Fraction ↔ Decimal Conversion

Conversion Rules

Fraction → Percentage : multiply by 100 | Percentage → Fraction : divide by 100 | Percentage → Decimal : divide by 100

Fraction	Decimal	Percentage
1/2	0.5	50%
1/4	0.25	25%
3/4	0.75	75%
1/5	0.2	20%
1/8	0.125	12.5%
1/3	0.333...	33.33%
2/3	0.666...	66.67%

1.5 Visual Explanation

Imagine a square grid of 100 small boxes (10 × 10). If you shade 25 of them, you have shaded 25% of the grid — this is exactly the same as shading 1/4 of the grid. This mental image is extremely useful for beginners: percentage is just "how many boxes out of 100 are shaded."

Another useful visual is the pie chart: a full circle represents 100%, and each slice's percentage tells you what fraction of the circle (360°) it occupies. To convert percentage to degrees in a pie chart: **degrees = (percentage/100) × 360**.

Section 2 — Complete Formula Sheet

2.1 Basic Percentage

Concept	Formula
x% of y	(x/100) × y
What % is A of B	(A/B) × 100
Finding the number when % is known	If x% of N = A, then N = (A × 100)/x

2.2 Percentage Increase / Decrease

Concept	Formula
% Increase	[(New − Original)/Original] × 100
% Decrease	[(Original − New)/Original] × 100
New value after x% increase	Original × (1 + x/100)
New value after x% decrease	Original × (1 − x/100)

2.3 Net / Successive Percentage Change

Two successive changes of a% and b%

Net % change = a + b + (ab/100)

Use a negative sign for decreases. E.g., +20% then −10% → a=20, b=−10.

2.4 Reverse Percentage

Situation	Formula
Value increased by r% to become F	Original = F / (1 + r/100)
Value decreased by r% to become F	Original = F / (1 − r/100)

2.5 Population Problems

Population after n years

$$P_n = P_0 \times (1 \pm r/100)^n$$

Use "+" for growth, "−" for decline. For population n years *ago*: $P_0 = P_n / (1 \pm r/100)^n$.

2.6 Salary / Income & Expenditure

Concept	Formula
Savings	Income − Expenditure
Savings %	(Savings/Income) × 100
New savings after income & expenditure change	New Income − New Expenditure

2.7 Marks / Exam Problems

Concept	Formula
Percentage scored	(Marks obtained / Maximum marks) × 100
Passing marks	(Pass % / 100) × Maximum marks
Marks obtained (fail by k)	Passing marks − k
Marks obtained (pass by k, above passing)	Passing marks + k

2.8 Election Problems

Concept	Formula
Valid votes	Total votes − Invalid votes
Winning margin (two candidates)	(Winner% − Loser%) × Valid votes / 100
Votes for a candidate	Candidate% × Valid votes / 100

2.9 Profit & Loss ↔ Percentage

Concept	Formula
Selling Price with profit%	SP = CP × (1 + Profit%/100)
Selling Price with loss%	SP = CP × (1 − Loss%/100)
Profit / Loss %	[(SP − CP)/CP] × 100

2.10 Discount

Concept	Formula
Selling Price	SP = MP × (1 − Discount%/100)
Successive discounts d1%, d2%	Single equivalent discount = d1 + d2 − (d1×d2)/100

2.11 Price Change vs Consumption/Expenditure

If price increases by r%, reduce consumption by

[r / (100 + r)] × 100 %, to keep expenditure constant

If price decreases by r%, consumption may increase by

[r / (100 − r)] × 100 %, to keep expenditure constant

Section 3 — Core Concepts

3.1 Basic Percentage Calculation

Definition: Expressing a quantity as parts per hundred of a whole.

> **$x\%$ of $y = (x \times y)/100$**

Intuition: Think of "of" as multiplication and "%" as "divide by 100." So "20% of 150" literally reads as $(20/100) \times 150$.

Shortcut: To find 10% of any number, just move the decimal point one place left. 1% = move two places left. Build all other percentages from these two.

> **Common Mistake:** Students often confuse "20% of 150" with "150% of 20" — both are numerically equal (30), but confusing the base quantity in worded problems leads to wrong setups in harder questions.

> **Interview Tip:** When asked to estimate mentally, always build from 10% and 1% — it is faster and less error-prone than direct multiplication.

3.2 Percentage Increase / Decrease

Definition: The relative change in a quantity, measured against its *original* value — never the new value.

> **% change = [(New − Original)/Original] × 100**

Intuition: The denominator is always the starting point, because "percentage change" answers the question "relative to where we started, how much did we move?"

Shortcut: If a value becomes k times the original, % increase = $(k - 1) \times 100$. E.g., value doubles ($k=2$) → 100% increase; value becomes 1.5× → 50% increase.

> **Common Mistake:** Using the new value as the denominator instead of the original value. Also, forgetting that a decrease from 100 to 80 (20% decrease) is NOT reversed by a 20% increase back to 100 — you actually need a 25% increase (80 → 100).

> **Interview Tip:** Always ask "increase/decrease with respect to what?" before setting up the formula — the reference value is the most common trap.

3.3 Successive Percentage Change

Definition: When a value undergoes two or more percentage changes one after another (not simultaneously).

> **Net % change = a + b + ab/100 (use negative values for decreases)**

Intuition: This formula is derived from multiplying the two multipliers: (1 + a/100)(1 + b/100) = 1 + a/100 + b/100 + ab/10000. Converting back to a percentage gives a + b + ab/100.

Shortcut: For quick multiple-choice elimination — if one change is an increase and the other an equal-magnitude decrease (e.g., +20%, −20%), the net effect is ALWAYS a net decrease equal to $(a^2)/100$, never zero.

> **Common Mistake:** Simply adding the two percentages (a + b) and ignoring the cross term ab/100 — this is only valid when one of the values is very small (used for rough estimation only).

> **Interview Tip:** For three successive changes a%, b%, c%, just multiply three factors: (1±a/100)(1±b/100)(1±c/100), then convert back to a percentage change. Don't try to memorize a three-term algebraic formula.

3.4 Reverse Percentage

Definition: Working backward from a final value to find the original value, given the percentage change applied.

> **Original = Final Value / (1 ± r/100)**

Intuition: If increasing the original by r% gives the final value, the final value already equals Original × (1+r/100). Dividing undoes the multiplication.

Shortcut: Convert the multiplier to a clean fraction. A 25% increase means final = (5/4) × original, so original = (4/5) × final.

> **Common Mistake:** Dividing the final value by (1 − r/100) when the change was actually an increase, or vice-versa. Always match the sign to the direction of change described in the question.

> **Interview Tip:** Reverse percentage almost always hides inside "successive change" and "population" word problems — learn to recognize the pattern "final value given, find original."

3.5 Population Growth & Decline

Definition: Population change over multiple years, where each year's growth/decline is applied to the *previous* year's population (compounding effect, similar to compound interest).

> $$P_n = P_0 (1 \pm r/100)^n$$

Intuition: Each year's percentage is applied on top of the already-changed population, not on the original — this is exactly how compound interest works.

Shortcut: If the rate differs each year (r1%, r2%, r3%...), simply multiply the individual factors: P_0 × (1±r1/100)(1±r2/100)(1±r3/100).

> **Common Mistake:** Treating population growth as simple (additive) percentage instead of compounding — this only works for a single year, not multiple years.

> **Interview Tip:** Population problems are compound-interest problems in disguise — if you're comfortable with CI, population questions become trivial.

3.6 Salary, Income & Expenditure

Definition: Problems relating income, spending, and savings, often with percentage changes to one or more of these quantities.

> **Savings = Income − Expenditure | Savings % = (Savings/Income) × 100**

Intuition: Since Savings is a *difference*, a small percentage change in Income or Expenditure can cause a disproportionately large percentage change in Savings — this is the classic placement trap.

Shortcut: Assume Income = 100 (or any convenient round number) when actual figures aren't given — this makes percentage calculations trivially easy.

> **Common Mistake:** Assuming that if income increases by x% and expenditure increases by y%, savings also increases by (x−y)%. This is FALSE — savings is a small residual, so its percentage change must be recalculated from actual values, not by subtracting percentages.

> **Interview Tip:** Always compute the actual old and new savings values explicitly before finding % change in savings — never shortcut this particular case.

3.7 Marks & Examination Problems

Definition: Percentage as applied to test scores, pass marks, and maximum marks.

> **% scored = (Marks obtained/Max marks) × 100**

Intuition: "Failing by k marks" means the student's score is exactly k less than the passing mark; "passing by k marks above the pass mark" means the score is k more than the passing mark.

Shortcut: Set up two equations — one for the passing mark in terms of maximum marks, one for the given student's score — and solve simultaneously. This handles almost all marks-based percentage problems.

Common Mistake: Confusing "marks obtained" with "passing marks" — read the question carefully to identify which quantity is being described.

Interview Tip: Whenever two scenarios are given (two students, or a student under two different passing criteria), always frame both as separate linear equations in the maximum marks — this is a very common TCS NQT pattern.

3.8 Election Problems

Definition: Percentage distribution of votes among candidates, factoring in invalid votes.

Winning Margin = (Winner% − Loser%) × Valid Votes / 100

Intuition: Only valid votes are distributed among candidates as percentages — invalid votes must be removed from the total first.

Shortcut: In a two-candidate race, if the winner gets w% of valid votes, the margin percentage is simply (2w − 100)%, so Valid Votes = Margin ÷ [(2w−100)/100].

Common Mistake: Applying the winning percentage to the total votes polled instead of only the valid votes when invalid votes are mentioned.

Interview Tip: Always identify whether the question gives "total votes polled" or "valid votes" — this distinction changes the entire setup.

3.9 Profit, Loss & Discount (Percentage View)

Definition: Cost Price (CP), Selling Price (SP), Marked Price (MP), and Discount, all linked through percentage relationships.

SP = MP(1 − D%/100) → SP = CP(1 + Profit%/100)

Intuition: Marked Price → Discount → Selling Price → compared with Cost Price → gives Profit/Loss. Each arrow is a percentage transformation.

Shortcut: Assume CP = 100 always. Compute MP and SP as percentages of 100, then read off the profit/loss % directly from the final number.

Common Mistake: Calculating discount % on Cost Price instead of Marked Price — discount is always applied on the Marked Price, not the Cost Price.

Interview Tip: "Marked 40% above cost, then 10% discount" is one of the most repeated TCS NQT patterns — always solve it as CP → MP (mark-up) → SP (discount) → compare with CP.

3.10 Price Change vs Consumption/Expenditure

Definition: When the price of a commodity changes, how much should consumption change to keep total expenditure (Price × Quantity) constant?

If price ↑ by r%, reduce consumption by [r/(100+r)]×100%

Intuition: Expenditure = Price × Quantity. If Price becomes (100+r)/100 times, Quantity must become 100/(100+r) times to keep the product the same — this ratio, converted to a % decrease, gives the formula.

Shortcut: Memorize the "inverse fraction trick" — new quantity multiplier is always the reciprocal of the new price multiplier when expenditure is fixed.

Common Mistake: Using the increase/decrease formula symmetrically (i.e., assuming a 25% price rise needs exactly a 25% consumption cut) — the correct reduction is always less than r%, per the formula above.

Interview Tip: This exact question type — "price increases by X%, find % reduction in consumption to keep expenditure same" — appears almost every year in TCS NQT and Capgemini papers.

Section 4 — Short Tricks & Mental Maths

4.1 Fraction to Percentage — Master Table (Memorize This!)

Fraction	%	Fraction	%
1/2	50%	1/9	11.11%
1/3	33.33%	1/10	10%
2/3	66.67%	1/11	9.09%
1/4	25%	1/12	8.33%
3/4	75%	1/13	7.69%
1/5	20%	1/14	7.14%
1/6	16.67%	1/15	6.67%
1/7	14.28%	1/16	6.25%
1/8	12.5%	1/20	5%

Why memorize this? Options in TCS NQT are frequently written as clean fractions (like 1/6, 1/7). Recognizing "16.67%" instantly as 1/6 saves 20–30 seconds per question.

4.2 Percentage to Fraction — Reverse Lookup

To convert any percentage to a fraction quickly: write it over 100 and simplify.

37.5% = 375/1000 = 3/8 | 62.5% = 625/1000 = 5/8

4.3 Mental Maths — Building Any Percentage from 10% and 1%

Target %	Trick
10%	Shift decimal one place left
1%	Shift decimal two places left
5%	Half of 10%
20%	Double of 10%
15%	10% + 5%
25%	Quarter of the number (divide by 4)

50%	Half of the number
2.5%	Quarter of 10%
12.5%	Eighth of the number (divide by 8)
33.33%	One-third of the number

4.4 Fast Calculation Tricks

- **Cross trick:** x% of y = y% of x. So 8% of 25 is hard to compute directly, but 25% of 8 = 2 instantly.
- **Percentage of a percentage:** "20% of 50% of 400" = 0.20 × 0.50 × 400 = 40. Multiply the decimals together first.
- **Doubling/halving:** To find 75%, take half + quarter of the number.
- **Successive change shortcut for equal ± values:** +x% then −x% always nets to a decrease of $x^2/100$ %.

4.5 Percentage Tables to Memorize (Squares Connection)

For quick successive-change calculations, memorizing squares up to 30 helps compute the ab/100 correction term instantly.

n	n^2	n	n^2
10	100	20	400
12	144	22	484
15	225	25	625
18	324	30	900

4.6 Important Values to Memorize Before the Exam

> **50%=1/2 25%=1/4 20%=1/5 12.5%=1/8 10%=1/10 33.33%=1/3 16.67%=1/6 6.25%=1/16 5%=1/20 4%=1/25**

Section 5 — 20 Solved Examples

Progression: Examples 1–5 (Easy) → 6–13 (Medium) → 14–20 (Difficult)

Example 1

EASY

Q: What is 15% of 240?

Approach: Direct application of x% of y = xy/100.

Solution: 15% of 240 = (15 × 240)/100 = 3600/100 = 36

Shortcut: 10% of 240 = 24; 5% = half of that = 12; 15% = 24+12 = 36

Final Answer: 36

Example 2

EASY

Q: 45 is what percent of 180?

Approach: Use (A/B) × 100.

Solution: (45/180) × 100 = 0.25 × 100 = 25%

Shortcut: 45/180 simplifies to 1/4 immediately, which is 25%.

Final Answer: 25%

Example 3

EASY

Q: Convert 3/8 into a percentage.

Approach: Multiply the fraction by 100.

Solution: (3/8) × 100 = 300/8 = 37.5%

Shortcut: 1/8 = 12.5% (memorized), so 3/8 = 3 × 12.5% = 37.5%

Final Answer: 37.5%

Example 4

EASY

Q: A quantity increases from 80 to 100. Find the percentage increase.

Approach: % increase = [(New−Original)/Original] × 100

Solution: [(100−80)/80] × 100 = (20/80) × 100 = 25%

Shortcut: 20 out of 80 is exactly 1/4, i.e., 25%.

Final Answer: 25%

Example 5

EASY

Q: A quantity decreases from 120 to 90. Find the percentage decrease.

Approach: % decrease = [(Original−New)/Original] × 100

Solution: [(120−90)/120] × 100 = (30/120) × 100 = 25%

Final Answer: 25%

Example 6

MEDIUM

Q: The price of an item increases by 20% and then decreases by 20%. Find the net percentage change.

Approach: Apply successive percentage formula with a=20, b=−20.

Solution: Net = a+b+ab/100 = 20−20+(20×−20)/100 = 0 − 4 = −4%

Shortcut: Equal magnitude +x then −x always gives a net decrease of $x^2/100$ = 400/100 = 4%.

Final Answer: 4% decrease

Example 7

MEDIUM

Q: A shopkeeper gives successive discounts of 10% and 20% on a marked price of Rs. 500. Find the final selling price.

Approach: Multiply successive discount factors.

Solution: SP = 500 × 0.9 × 0.8 = 500 × 0.72 = Rs. 360

Shortcut: Single equivalent discount = 10+20−(10×20)/100 = 30−2 = 28%; SP = 500×0.72 = 360.

Final Answer: Rs. 360

Example 8

MEDIUM

Q: A's salary is increased by 25% and then decreased by 20%. Find the net percentage change.

Approach: Successive change formula, a=25, b=−20.

Solution: Net = 25 − 20 + (25×−20)/100 = 5 − 5 = 0%

Final Answer: No change (0%)

Example 9

MEDIUM

Q: In an election between two candidates, the winner gets 60% of valid votes and wins by 200 votes. Find the total valid votes.

Approach: Margin % = 60−40 = 20%, equals 200 votes.

Solution: 20% of V = 200 → V = 200×100/20 = 1000

Final Answer: 1000 valid votes

Example 10

MEDIUM

Q: The population of a town is 40,000. It increases by 10% in the first year and decreases by 10% in the second year. Find the population after 2 years.

Approach: Apply compounding formula with r1=+10%, r2=−10%.

Solution: 40000 × 1.1 × 0.9 = 40000 × 0.99 = 39,600

Shortcut: +10% then −10% always nets to −1% (10^2/100).

Final Answer: 39,600

Example 11

MEDIUM

Q: A man spends 75% of his income and saves the remaining Rs. 3000. Find his income.

Approach: Savings % = 100−75 = 25%; set up equation.

Solution: 25% of Income = 3000 → Income = 3000×100/25 = Rs. 12,000

Final Answer: Rs. 12,000

Example 12

MEDIUM

Q: In an exam, the passing percentage is 40%. A student scored 150 marks and failed by 30 marks. Find the maximum marks.

Approach: Passing marks = 150+30 = 180 = 40% of max marks.

Solution: Max marks = 180×100/40 = 450

Final Answer: 450 marks

Example 13

MEDIUM

Q: The price of sugar increases by 25%. By what percentage should a household reduce consumption to keep expenditure unchanged?

Approach: Use the price-consumption formula: r/(100+r) × 100.

Solution: 25/(100+25) × 100 = 25/125 × 100 = 20%

Final Answer: 20% reduction

Example 14

DIFFICULT

Q: Two numbers are respectively 20% and 50% more than a third number. Find the ratio of the first number to the second.

Approach: Let the third number = 100; find first and second in terms of it.

Solution: First = 120, Second = 150. Ratio = 120:150 = 4:5

Final Answer: 4 : 5

Example 15

DIFFICULT

Q: A's income is 60% more than B's income. By what percentage is B's income less than A's?

Approach: Let B=100, so A=160. Difference relative to A.

Solution: (60/160) × 100 = 37.5%

Shortcut: If A is x% more than B, B is less than A by [x/(100+x)]×100%.

Final Answer: 37.5%

Example 16

DIFFICULT

Q: In a three-candidate election, votes are distributed in ratio 4:3:2 with total valid votes 90,000. Find the winning margin (votes) between the winner and runner-up.

Approach: Divide total votes into 9 parts.

Solution: 1 part = 90000/9 = 10,000. Winner = 40,000; Runner-up = 30,000. Margin = 10,000

Final Answer: 10,000 votes

Example 17

DIFFICULT

Q: A trader marks his goods 40% above the cost price and then gives a 10% discount. Find his profit percentage.

Approach: Assume CP=100; find MP, then SP; compare to CP.

Solution: MP = 140; SP = 140×0.9 = 126; Profit% = 26%

Final Answer: 26% profit

Example 18

DIFFICULT

Q: The population of a village decreases by 10% every year due to migration. If the present population is 72,900, find the population 3 years ago.

Approach: Reverse-apply compounding: $P_0 = P_n/(1-r/100)^n$

Solution: $P_0 = 72900/(0.9)^3 = 72900/0.729 = 1{,}00{,}000$

Final Answer: 1,00,000

Example 19

DIFFICULT

Q: In a class, 60% of the students are boys. 20% of the boys and 25% of the girls fail an exam. Find the overall pass percentage of the class.

Approach: Assume total students = 100; compute failures separately for boys and girls.

Solution: Boys=60, Girls=40. Boys failed=20%×60=12. Girls failed=25%×40=10. Total failed=22. Pass%=100−22=78%

Final Answer: 78%

Example 20

DIFFICULT

Q: A shopkeeper marks an item 100% above the cost price, then offers successive discounts of 25% and 20%. Find his overall profit or loss percentage.

Approach: Assume CP=100; compute MP, apply both discounts sequentially, compare final SP with CP.

Solution: MP = 200. After 25% discount: 200×0.75=150. After 20% discount: 150×0.8=120. Since CP=100 and SP=120, Profit%=20%

Shortcut: Combined multiplier = 2 × 0.75 × 0.8 = 1.2 → 20% profit directly.

Final Answer: 20% profit

Section 6 — Previous-Year Frequently Tested Concept Patterns

Based on recurring patterns across TCS NQT and similar service-based company placement exams, the following percentage sub-topics appear most frequently. Focus your revision time here.

6.1 Successive Percentage Change

Almost every paper includes at least one "increase then decrease" or "two discounts" style question. Master the a+b+ab/100 formula and its three-term extension.

6.2 Salary Increment / Decrement Chains

Multi-step salary revision problems (hike, then cut, then bonus%) are extremely common — treat each step as a multiplier.

6.3 Population Growth/Decline (Compound Style)

Frequently tested with different rates for different years, or asking for population "n years ago" (reverse percentage + compounding combined).

6.4 Election Vote-Share Problems

Watch for the distinction between "total votes" and "valid votes" — invalid vote percentage is a common twist.

6.5 Profit & Loss with Marked Price and Discount

The "mark up X% above CP, then discount Y%" pattern is one of the single most repeated question types across all service-based company exams.

6.6 Discount Chains (Successive Discounts)

Multiple discounts applied one after another — always solve via multiplier chains, never by simple addition.

6.7 Income & Expenditure / Savings

Questions where income and expenditure both change by different percentages, and you must find the % change in savings — a classic trap topic.

6.8 Marks & Pass/Fail Threshold Problems

Two-student or two-scenario marks problems solved via simultaneous linear equations in maximum marks.

6.9 Reverse Percentage ("Find the Original")

Given a final value after one or more percentage changes, find the starting value — tests conceptual understanding rather than formula recall.

6.10 Mixed / Combination Percentage Problems

Higher-difficulty questions blend two or more of the above patterns in a single question (e.g., profit% + successive discount, or population + reverse percentage) — these appear in the later, harder sections of the paper.

> **Exam Strategy:** If you can confidently solve one representative question from each of the 10 patterns above, you can handle roughly 90% of all percentage-based questions that appear in TCS NQT and similar exams.

Section 7 — Classroom Practice Questions

30 Original Questions — 10 Medium (Q1–Q10) + 20 Difficult (Q11–Q30). No hints or solutions provided — attempt independently, then check the Answer Key in Section 8.

Medium (Q1 – Q10)

Q1. What is 18% of 250?

Q2. 63 is what percent of 420?

Q3. Express 7/16 as a percentage.

Q4. The price of an article is increased from Rs. 600 to Rs. 750. Find the percentage increase.

Q5. A number is decreased by 30% to give 84. Find the original number.

Q6. In an election between two candidates, the winner secured 55% of the votes and won by 600 votes. Find the total votes polled.

Q7. A shopkeeper offers a discount of 15% on a marked price of Rs. 800. Find the selling price.

Q8. Ravi's monthly salary is Rs. 40,000. He spends 65% of it. Find his monthly savings.

Q9. If A's income is 25% more than B's income, by what percentage is B's income less than A's?

Q10. The population of a town increases by 8% annually. If the current population is 25,000, find the population next year.

Difficult (Q11 – Q30)

Q11. Two successive discounts of 20% and 25% are equivalent to what single discount?

Q12. A number is increased by 20% and then decreased by 20%. Find the net percentage change.

Q13. An employee's salary is first increased by 10% and then decreased by 10%. If his final salary is Rs. 19,800, find his original salary.

Q14. A candidate who scores 35% of the maximum marks fails by 40 marks, while another candidate who scores 55% gets 60 marks more than the passing marks. Find the maximum marks.

Q15. The price of a commodity is increased by 30%. By what percentage must a family reduce its consumption to keep expenditure unchanged?

Q16. The population of a city is 84,000. It increases by 20% in the first year and decreases by 10% in the second year. Find the population after 2 years.

Q17. A man spends 80% of his income. His income increases by 20% and his expenditure increases by 10%. Find the percentage increase in his savings.

Q18. In an election, 10% of the votes were declared invalid. The winning candidate secured 60% of the valid votes and won by 9000 votes. Find the total number of votes polled.

Q19. A trader marks his goods 50% above cost price and allows a discount of 20%. Find his profit percentage.

Q20. Number A is 30% less than number B. By what percentage is B more than A?

Q21. If the numerator of a fraction is increased by 20% and the denominator is decreased by 10%, the resulting fraction is 4/5. Find the original fraction.

Q22. A's salary is 20% more than B's salary, and B's salary is 25% more than C's salary. If C earns Rs. 20,000, find A's salary.

Q23. In a college, 40% of the students are girls. If 25% of boys and 10% of girls play cricket, what percentage of the total students play cricket?

Q24. A machine depreciates at 10% per annum. If its present value is Rs. 72,900, what was its value 2 years ago?

Q25. A shopkeeper sells an article at a 10% loss. Had he sold it for Rs. 108 more, he would have gained 8%. Find the cost price.

Q26. Ram's income is 25% less than Shyam's income. By what percentage is Shyam's income more than Ram's?

Q27. In an examination, 70% of students passed in Maths and 65% passed in Science, while 50% passed in both subjects. Find the percentage of students who failed in both subjects.

Q28. A sum of money increases by 44% in 2 years under compound interest, compounded annually. Find the annual rate of interest.

Q29. The length of a rectangle is increased by 25% and its breadth is decreased by 20%. Find the percentage change in its area.

Q30. In a factory, 5% of the total production of 8,000 units is defective. If 60% of the defective units are rejected, how many defective units are accepted (not rejected)?

Section 8 — Answer Key

Q1 - 45
Q2 - 15%
Q3 - 43.75%
Q4 - 25%
Q5 - 120
Q6 - 6000
Q7 - Rs. 680
Q8 - Rs. 14,000
Q9 - 20%
Q10 - 27,000
Q11 - 40%
Q12 - 4% decrease
Q13 - Rs. 20,000
Q14 - 500 marks
Q15 - 23.08%
Q16 - 90,720
Q17 - 60%
Q18 - 50,000
Q19 - 20%
Q20 - 42.86%
Q21 - 3/5
Q22 - Rs. 30,000
Q23 - 19%
Q24 - Rs. 90,000
Q25 - Rs. 600
Q26 - 33.33%
Q27 - 15%
Q28 - 20%
Q29 - No change (0%)
Q30 - 160 units

Section 9 — One-Page Revision Sheet

Core Formulas

x% of y = xy/100

% change = [(New−Original)/Original]×100

Successive change: a+b+ab/100

Reverse %: Original = Final/(1±r/100)

Population: $P_n = P_0(1±r/100)^n$

SP = MP(1−D%/100); SP = CP(1+Profit%/100)

Price↑r% → reduce consumption by r/(100+r)×100%

Winning margin = (Winner%−Loser%)×Valid votes/100

Key Fraction–Percentage Values

1/2=50% · 1/3=33.33% · 1/4=25% · 1/5=20% · 1/6=16.67% · 1/7=14.28% · 1/8=12.5% · 1/9=11.11% · 1/10=10% · 1/11=9.09% · 1/12=8.33% · 1/16=6.25% · 1/20=5% · 1/25=4%

Mental Maths Building Blocks

10% → shift decimal left once. 1% → shift decimal left twice. 5% = half of 10%. 25% = ÷4. 50% = ÷2. 12.5% = ÷8. 33.33% = ÷3.

Frequently Tested Patterns

1. Successive % change 2. Salary hike/cut chains 3. Population growth/decline 4. Election vote-share 5. Profit/Loss with markup & discount 6. Successive discounts 7. Income-Expenditure-Savings 8. Marks/Pass-fail 9. Reverse percentage 10. Mixed combination problems

Common Mistakes to Avoid

- Using new value instead of original as the base for % change
- Adding successive percentages directly without the ab/100 correction
- Applying discount % on Cost Price instead of Marked Price
- Assuming savings % change equals (income% − expenditure%)
- Using total votes instead of valid votes in election problems
- Treating population growth as simple (additive) instead of compounding

Last-Minute Reminders

- +x% then −x% always nets to a decrease of x^2/100 %
- If A is x% more than B, B is less than A by [x/(100+x)]×100%
- Assume base value = 100 whenever actual figures aren't given
- For 3+ successive changes, multiply factors — don't extend the 2-term formula
- Read carefully: "of" = original base; "more/less than" = reference value